INTRODUCTION TO INFORMATION GEOMETRY

Stability, Parameters, and Inference

Week 2 · Lecture 2

Christopher Lee

The derivative at an interior state

Write $\Delta(x) = \alpha + \beta x$, so

$$F(x) = x(1-x)(\alpha + \beta x).$$

At an interior equilibrium x^*, the factor $\Delta(x^*)$ vanishes. Differentiation gives

$$F'(x^*) = \beta x^*(1-x^*).$$

The slope of the advantage line determines local attraction or repulsion.

The local rate of coexistence

Take the familiar model

$$F(x) = x(1-x)(3-4x), \qquad x^* = \tfrac{3}{4}.$$

Find the linearization coefficient:

$$F'(x^*) = ?.$$

Is the equilibrium attracting or repelling?

The local rate of coexistence

Take the familiar model

$$F(x) = x(1-x)(3-4x), \qquad x^* = \tfrac{3}{4}.$$

The advantage has slope $\beta = -4$, so

$$F'(x^*) = -4 \cdot \frac{3}{4} \cdot \frac{1}{4} = -\frac{3}{4}.$$

The equilibrium is attracting, with local decay $u(t) \approx u(0)e^{-3t/4}$.

When linearization becomes inconclusive

For $F(x) = 3x(1-x)^2$, the endpoint $x = 1$ attracts from the interior.

$$F'(1) = 0.$$

Set $u = 1 - x$. Near $u = 0$,

$$u' = -3(1-u)u^2 \approx -3u^2.$$

The displacement decays approximately like $1/(3t + C)$. The phase line settles the stability question.

Changing one entry of a payoff matrix

Consider the family

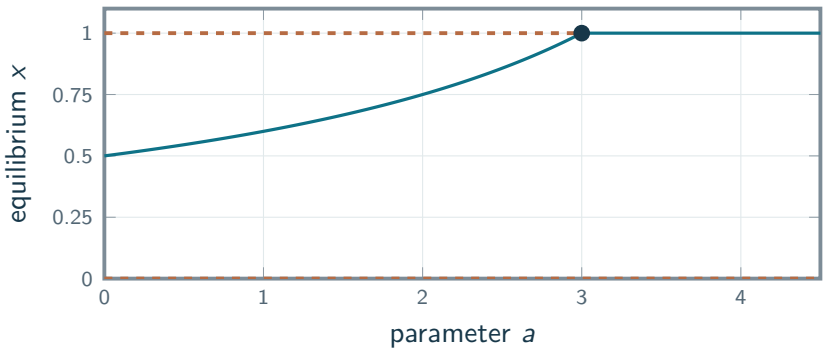
$$M(a) = \begin{pmatrix} a & 4 \\ 3 & 1 \end{pmatrix}, \qquad \Delta_a(x) = 3 + (a-6)x.$$

For $a < 3$, the interior equilibrium is

$$x^*(a) = \frac{3}{6-a}.$$

It approaches $x = 1$ as a approaches 3 from below.

A boundary bifurcation



At $a = 3$, the attracting interior branch meets the boundary.

Payoff transformations

Only the difference $f_A - f_B$ enters the equation for x. If $\mathbf{1} = (1,1)^T$ and $r = (r_A, r_B)^T$, then

$$\widetilde{M} = M + \mathbf{1} r^T$$

adds the same opponent-dependent term $r_A x + r_B(1-x)$ to both fitnesses. It changes their common growth level while preserving composition dynamics.

Equivalent interaction matrices

Begin with $M = \begin{pmatrix} 2 & 4 \\ 3 & 1 \end{pmatrix}$. Subtract the second row from both rows, column by column.

$$\widetilde{M} = \begin{pmatrix} ? & ? \\ ? & ? \end{pmatrix}.$$

What is the resulting advantage $\widetilde{f}_A - \widetilde{f}_B$?

Equivalent interaction matrices

Begin with $M = \begin{pmatrix} 2 & 4 \\ 3 & 1 \end{pmatrix}$. Subtract the second row from both rows, column by column.

$$\widetilde{M} = \begin{pmatrix} -1 & 3 \\ 0 & 0 \end{pmatrix}.$$

Both matrices yield $\widetilde{f}_A - \widetilde{f}_B = 3 - 4x$.

Multiplying the payoffs

If every payoff is multiplied by $\lambda > 0$, then

$$\widetilde{F}(x) = \lambda F(x).$$

The same orbit is traversed at a different speed:

$$\widetilde{x}(t) = x(\lambda t).$$

The phase-line arrows and equilibrium positions stay the same.

Designing a desired equilibrium

We want an attracting composition $q = 0.4$ with local rate $u' = -0.6u$. Choose

$\Delta(x) = b(q - x)$.

$$F'(q) = -bq(1 - q) = -0.6.$$

Find b and $\Delta(x)$.

Designing a desired equilibrium

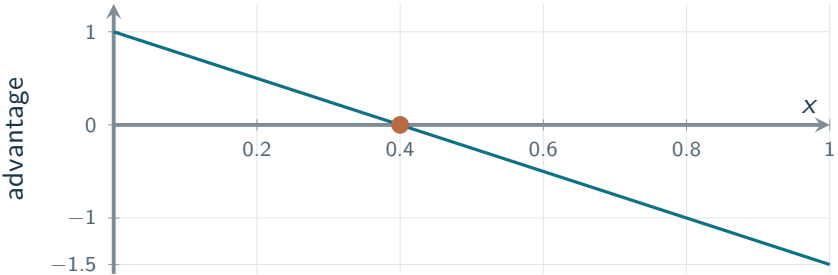
We want an attracting composition $q = 0.4$ with local rate $u' = -0.6u$. Choose

$\Delta(x) = b(q - x)$.

$$b = \frac{0.6}{(0.4)(0.6)} = 2.5.$$

Then $\Delta(x) = 1 - 2.5x$.

A phase portrait constructed to order



The zero specifies the target state. The slope controls its local return rate.

Recovering an advantage from data

Measurements of instantaneous velocity give

$$(x, x') = (0.2, 0.16), \qquad (0.6, -0.24).$$

If $\Delta(x) = \alpha + \beta x$, find α and β using

$$\Delta(x) = \frac{x'}{x(1-x)}.$$

Recovering an advantage from data

Measurements of instantaneous velocity give

$$(x, x') = (0.2, 0.16), \qquad (0.6, -0.24).$$

The corresponding advantages are 1 and -1. Thus

$$\Delta(x) = 2 - 5x, \qquad x^* = \frac{2}{5}.$$

Inference near a pure population

The inferred advantage divides by $x(1-x)$.

$$\Delta(x) = \frac{x'}{x(1-x)}.$$

A small error ε in the measured velocity changes the advantage estimate by

$$\frac{\varepsilon}{x(1-x)}.$$

For rare strategies, the same measurement error has a larger effect.